Thambawita Maddumage Sajith Priyankara

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Research Interests

Functional data analysis, object-oriented data analysis, statistical analysis on manifolds,high-dimensional statistical modeling, protein–ligand binding analysis, statistical methods for spatial transcriptomics and genomics, longitudinal and epidemiologic data analysis.

Education

Ph.D. Candidate in Mathematics Concentration in Statistics 2021–Present
Texas Tech University, Lubbock, TX

M.S. in Statistics 2021–2023
Texas Tech University, Lubbock, TX

B.Sc. in Statistics 2015–2019
University of Peradeniya, Sri Lanka

Research Experience

Ph.D. Researcher : Statistical Modeling of Protein–Ligand Binding Sites 2021–Present
Department of Mathematics & Statistics, Texas Tech University

- Develop statistical representations of protein binding sites using functional data analysis to characterize structural and elemental variation in protein–ligand binding sites.
- Apply geometry-aware methods based on constrained functions, covariance structures, and statistical analysis on nonlinear spaces for comparing protein binding sites.
- Developed predictive models using statistical learning, machine learning, and deep learning approaches for ligand-binding classification and protein pocket matching, including random forests, penalized regression, and Siamese neural networks.
- Developed **FUSED**, a functional representation for joint structural and elemental analysis of protein–ligand binding sites; first-author manuscript **accepted for publication in the *Journal of Chemical Information and Modeling***.
- First-author manuscript, *Element-Specific Axis Distance Density (ESADD) Representations of Protein Binding Sites*, currently in final preparation.

Collaborative Biostatistics Researcher 2026–Present
Dr. Dawei Li Lab, Texas Tech University Health Sciences Center

- Collaborate with biomedical researchers on statistical analyses of alcohol use disorder, psychiatric comorbidities, early-life exposures, and related population health outcomes.
- Develop multivariable logistic regression, mixed-effects models, interaction analyses, multiple-imputation workflows, and sensitivity analyses.
- Contribute to statistical study design, covariate and confounder selection, and preparation of manuscripts.
- Second-author manuscript on alcohol use disorder and associated early-life exposures is currently in final preparation.
- Conduct computational research in spatial transcriptomics, developing geometry-aware statistical repre-

sentations to characterize spatial gene-expression patterns and tissue-level heterogeneity in Alzheimer's disease.

Researcher: Air Pollution and Respiratory Health 2019–2021
National Institute of Fundamental Studies, Sri Lanka

- Investigated associations between ambient particulate matter ($PM_{2.5}$ and PM_{10}) and respiratory hospitalizations using generalized additive models with quasi-Poisson regression.
- Evaluated lagged exposure effects and conducted age and sex-stratified analyses to identify populations at elevated risk.
- Led the statistical analysis and published a **first-author peer-reviewed paper**.

Researcher : Parental BMI Trajectories and Offspring Asthma 2020–2021
National Institute of Fundamental Studies, Sri Lanka

- Applied group-based trajectory modeling to Tasmanian Longitudinal Health Study data and used multinomial logistic regression to assess associations with offspring asthma outcomes.
- Identified exposure-specific potential confounders and developed adjusted statistical models.
- Co-authored a **peer-reviewed publication**.

Publications and Accepted Manuscripts

1. **Priyankara, S.**, Ellingson, L. (2026). FUSED: A Functional Representation for Joint Structural and Elemental Analysis of Protein Ligand Binding Sites. *Journal of Chemical Information and Modeling.* **Accepted for publication.** bioRxiv preprint. **First author.**
2. **Priyankara, S.**, et al. (2021). Ambient $PM_{2.5}$ and PM_{10} Exposure and Respiratory Disease Hospitalization in Kandy, Sri Lanka. *International Journal of Environmental Research and Public Health.* **First author.**
3. Bowatte, G., Bui, D. S., **Priyankara, S.**, et al. (2022). Parental preconception BMI trajectories from childhood to adolescence and asthma in the future offspring. *Journal of Allergy and Clinical Immunology.*

Manuscripts in Preparation

1. **Priyankara, S.**, Ellingson, L. Element-Specific Axis Distance Density (ESADD) Representations of Protein Binding Sites. *Manuscript in preparation.*
2. Manuscript on alcohol use disorder and associated early-life exposures. *Manuscript in final preparation; second author.*

Conference Presentations

1. **Priyankara, S.**, Ellingson, L. (2024). *Analysis of Covariance Matrix Distances in Protein Binding Sites.* Conference of Texas Statisticians.
2. **Priyankara, S.**, et al. (2021). *PM2.5 Exposure and Respiratory Disease Hospitalization in Sri Lanka.* Lung Science Conference – European Respiratory Society.
3. Priyankara, S., Abeysundara, S.P. (2020). *Early Recognition of Premature Birth.* International Conf. on Environmental and Medical Statistics.

Professional Experience

Graduate Part Time Instructor (Instructor of Record) 2022–Present
Department of Mathematics & Statistics, Texas Tech University

- Responsible for full course delivery, exam preparation, grading, and student support for 80–120 undergraduate students per section. Courses taught:
  - MATH 2300 – Statistical Methods
  - MATH 1330 – Introductory Mathematical Analysis I
  - MATH 1331 – Introductory Mathematical Analysis II

Graduate Teaching Assistant 2021–2022
Department of Mathematics & Statistics, Texas Tech University

Research Assistant 2019–2021
National Institute of Fundamental Studies, Sri Lanka

Technical Skills

Languages:	R, Python, STATA, MATLAB, C
Statistical Methods:	Functional data analysis, object-oriented data analysis, regression modeling (linear, logistic, multinomial), longitudinal data analysis, generalized additive models (GAM), multivariate statistics, time series analysis, survival analysis
Computational Methods:	Statistical modeling of protein structures, geometry-aware modeling, Riemannian geometry, high-dimensional data analysis,machine learning
Domain:	Ligand-protein binding prediction, spatial transcriptomics,epidemiology

Professional Memberships and Service

- Member, American Statistical Association.
- President, Statistical Circle, University of Peradeniya, 2017–2018.
- Volunteer, Refugee Services, Lubbock, Texas, 2023.
- Committee Member, Sri Lankan Students Association, Texas Tech University, 2022–2023

References

Dr. Leif Ellingson
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Relationship: Ph.D. Advisor

Dr. Alexandre Trindade
Professor, Department of Mathematics & Statistics
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Relationship: Ph.D. Committee Member

Dr. Dawei Li
Associate Professor, School of Medicine
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Email: Dawei.Li@ttuhsc.edu
Relationship: Research Collaborator